# Master IARFID
# Reconocimiento de Escritura (RES)
# Handwritten Text Recognition

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# Outline

- Handwritten Text Recognition Problem
- Program of the HTR (RES) course
- Evaluation

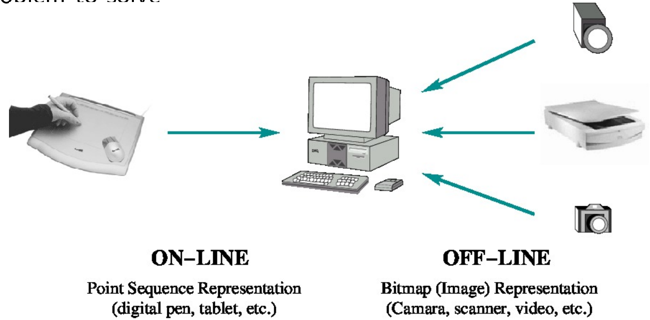
# Handwritten Text Recognition Problem

1. Still large part of the currently handled information can be found in handwritten documents:
   - Repositories of historical manuscripts and digital libraries in general
   - personal letters
   - handwritten notes
   - survey forms
   - ...
2. HTR has become an important issue for many industrial applications:
   - Recognition of legal amounts handwritten in bank checks
   - Signature verification
   - Reading of postal addresses
   - Indexing of handwritten document images
   - Civil and parish records
   - ...

# Handwritten Text Recognition Problem

1. In last decades, loads of documents have been digitized.
   - That allows its preservation and access.
   - Those documents holds unknown aspects of our political, social, economic and scientific history.
   - Unfortunately, the richness of these documents belongs unavailable.

2. The problem to solve



# Handwriting Text Recognition - Master IARFID

- Part 0 Introduction to the handwriting recognition problem.
- Part I Acquisition and preprocessing of text images.
- Part II Cursive handwriting text recognition
- Part III Applications based on HTR technology.

# Handwriting Text Recognition - Master IARFID

Part 0 Introduction to the handwriting recognition problem

Part I Acquisition and preprocessing of text images

- I.1 Document Image Processing.
  - I.1.1 Introduction & motivation
  - I.1.2 Image acquisition, analysis and enhancement
- I.2 Layout analysis: text block and line detection and extraction
  - I.2.1 Geometric normalizations: skew, slope, slant, warping and size
- I.3 Practical session: From page images to line images

# Handwriting Text Recognition - Master IARFID

Part II Modelling and decoding

- II.1 Introduction to (cursive) on/off-line
- II.2 Feature extraction for on/off-line
- II.3 Modelling and decoding
- II.4 Practical HTR session.

Part III - Applications based on HTR technology

- III.1 Word graphs
- III.2 Computer Assisted Transcription of Text Images: Catti
  - http://cat.prhlt.upv.es/iht
- III.3 Indexing and searching
  - http://transcriptorium.eu/demots/KWSdemos

# Dr. Moisés Pastor

Professor responsible for RES. Professor at Universitat Polècnica de València and Researcher at the Pattern Recognition and Human Language Technology research center.

- Research field
  - Pattern Recognition
  - Multimodal Interaction
  - Handwritten Text Recognition/Transcription
  - Computer Vision
- Publications: `https://www.prhlt.upv.es/u/mpastorg`

# Evaluation

- Oral presentation 50%
  - Evaluated by your colleagues using a standard form. https://poliformat.upv.es/access/content/group/DOC_33978_2019/Rubrica.pdf
- Academic works 50%